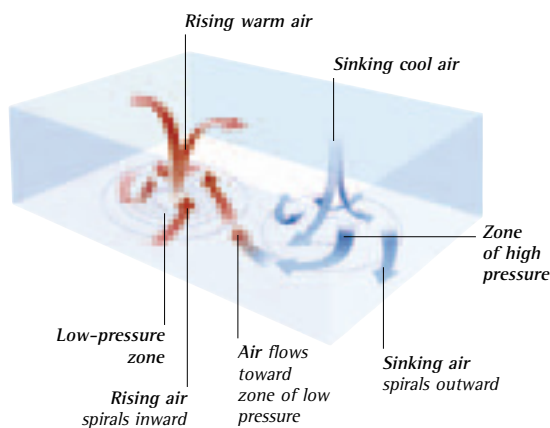


CYCLONES AND HURRICANES

Where a warm, moist air mass moving off an ocean meets a cold, dense air mass, the warm air rises above the cold air. This creates a swirling zone of rising air called a cyclone or depression. As the warm air rises it cools, so its moisture turns into clouds and rain. In temperate regions, prevailing winds carry these cyclones eastward, causing wet, windy weather. In the tropics, intense heat can generate the violent storms known as tropical cyclones, typhoons, or hurricanes.



◀ HIGHS AND LOWS

Cool air is denser and heavier than warm air, so it tends to sink, pushing down to create a zone of high atmospheric pressure. It spirals outward as it sinks, swirling clockwise north of the equator, and counterclockwise in the south. It flows toward low-pressure zones, where warmer air is rising, and spirals inward in the opposite direction. As it rises, the water vapor in the air turns to clouds and rain.



▲ PRESSURE AND WIND

The greater the pressure difference between nearby zones of high and low atmospheric pressure, the faster the air flows from one to the other. This causes strong winds, especially around low-pressure zones. They blow around the cores of these cyclones, often against the prevailing wind, causing the storms that lash cool ocean regions like the North Atlantic.

▲ SPIRALING CYCLONES

The weather in cooler oceans is dominated by spiraling low-pressure systems, or cyclones. Many form along the polar front where warm, moist tropical air meets colder, drier air from the polar regions. They move steadily eastward, carrying wind and rain with them. Similar but more intense cyclones develop over warmer oceans, like the Caribbean storm seen in this satellite image.

